

GREYNOC

PUBLIC CYBER HYGIENE GUIDE

Turn off NetBIOS and close unused ports

Version 1.3 - GreyNOC Research Bumped Working Release

Document ID	GN-CYH-NETBIOS-PORTS-001
Classification	PUBLIC / APPROVED FOR EXTERNAL RELEASE - BUMPED REVIEW COPY
Version	v1.3
Effective Date	May 4, 2026
Revision Date	June 12, 2026
Authors	GreyNOC Research
Document Owner	GreyNOC Cybersecurity Operations
Applies To	Windows endpoint hardening, small business networks, home labs, software test systems, AI-assisted development workstations, and managed endpoint rollouts
Public Release Rule	This document is informational cyber hygiene guidance. Replace environment-specific assumptions, validate changes in a pilot group, obtain authorization before testing or changing systems, and publish only through an approved GreyNOC channel.
Template Basis	GreyNOC Public Document Template GN-TPL-PUB-001

Release status	Bumped version prepared from v1.2 with added practical tips, safer command examples, exception handling, endpoint-management notes, and AI/dev exposure review improvements. Final publication should be performed only after owner approval is recorded.
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Document Control

This page records ownership, review state, and release history so the guide can be published, archived, and updated consistently.

Version	Date	Author	Summary of change
v1.0	May 4, 2026	GreyNOC	Initial public guide covering NetBIOS over TCP/IP, unused port discovery, firewall commands, validation, rollback, and public-use disclaimer.
v1.1	May 4, 2026	GreyNOC	Expanded each section for CEO release, added audience guidance, risk framing, pilot rollout steps, safer decision logic, stronger validation, and CEO approval language.
v1.2	May 4, 2026	GreyNOC Research	Converted to the GreyNOC public-facing document template style, updated authorship, and added AI-assisted development/test port review guidance including TCP 8000 and related local test services.
v1.3	June 12, 2026	GreyNOC Research	Bumped release with quick-win triage, common mistakes, improved PowerShell inventory, remote-address firewall examples, localhost binding tips, exception-review guidance, adjacent protocol notes, and expanded release checklist.

Review area	Release requirement
Communications	Approved GreyNOC wording, public-safe claims, clean formatting, and no customer-specific references.
Security	No exploit instructions, internal toolchain details, credentials, live IPs, customer environments, sensitive diagrams, or commands that encourage unauthorized testing.
Legal / ownership	Publication owner confirms the document is suitable for external release and archived with source, final PDF, approval notes, and checksum.

Table of Contents

- 01 Executive Summary
- 02 Scope and Audience
- 03 Why NetBIOS and Unused Ports Matter
- 04 Before You Make Changes
- 05 Turn Off NetBIOS on a Single Windows Device
- 06 Turn Off NetBIOS Using DHCP in Managed Networks
- 07 Find Listening Ports Before Closing Anything
- 08 Close Unused Ports Safely
- 09 Ports That Deserve Extra Review
- 10 AI-Assisted Development and Software Test Ports
- 11 AI/Test Port Review Workflow
- 12 Quick Wins, Tips, and Common Mistakes
- 13 Validation, Rollback, and Release Checklist
- 14 Public-Use Disclaimer and Publication Notes
- Appendix A - References

Reader action	Use this guide as a review and hardening workflow, not as a universal block list. Every port must be validated against business need before being disabled, blocked, or restricted.
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01 Executive Summary

NetBIOS over TCP/IP and unnecessary listening ports are common sources of avoidable network exposure. Modern Windows environments usually rely on DNS and Active Directory for name resolution, while many workstations, lab machines, and developer systems accumulate local services that are no longer needed or were started temporarily during testing.

GreyNOC Research recommends a measured approach: inventory first, pilot the change, validate business functions, document exceptions, and then roll out through controlled management. The goal is not to block every familiar port. The goal is to understand which services are listening, which are reachable, which are required, and which should be disabled or restricted.

v1.3 update

This bumped release adds field-ready tips: baseline exports, process-to-service mapping, remote-address restrictions, localhost binding examples, exception expiration, adjacent protocol review notes, and common mistakes that cause outages.

Key takeaways

- NetBIOS-related ports UDP 137, UDP 138, and TCP 139 should be reviewed and disabled or restricted where legacy dependency is not present.
- SMB over TCP 445 is related to Windows file sharing review, but it is not NetBIOS and must not be blocked blindly on systems that require file sharing, domain services, or administrative workflows.
- Ports opened by local software testing, AI tools, notebooks, and demo applications should be bound to localhost where possible and blocked from untrusted networks unless there is an approved business reason.
- Firewall blocking helps, but removing or disabling an unnecessary service is usually stronger because the service is no longer available to be exposed later.
- Use time-boxed exceptions. Any open management, file-sharing, or test port should have an owner, reason, source scope, review date, and rollback plan.

02 Scope and Audience

This guide is written for small business administrators, managed service providers, home lab operators, SOC analysts, help desk technicians, and software teams using AI-assisted testing tools. It focuses on practical Windows network hardening and local development exposure reduction.

In scope	Out of scope
Windows endpoint and server review; NetBIOS over TCP/IP; local listening port inventory; Windows Firewall rule review; pilot rollout; rollback planning; AI/dev/test port exposure review; endpoint-management rollout planning.	Unauthorized scanning; exploitation; bypassing controls; customer-specific architectures; live IP addresses; credentials; vendor-sensitive implementation details; guaranteed compliance claims.

Authorization boundary	Only scan, test, or change systems you own, administer, or are explicitly authorized to assess. Public cyber hygiene guidance must never be used as a reason to probe third-party networks.
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Recommended users of this document should have basic familiarity with Windows administration, PowerShell, network ports, and change control. Less experienced readers should test only in a lab or under supervision.

03 Why NetBIOS and Unused Ports Matter

NetBIOS over TCP/IP is a legacy name resolution and session service technology. In many modern Windows environments, DNS and Active Directory handle the work that NetBIOS historically supported. Leaving legacy discovery protocols and unnecessary listening services exposed can make systems easier to enumerate and can create avoidable paths for lateral movement.

This does not mean every NetBIOS-related setting or port should be blocked blindly. Some older applications, printers, file-sharing workflows, backup agents, or mixed-version networks may still depend on legacy behavior. The safest approach is measured exposure reduction: identify what is actually needed, disable what is not needed, and restrict what must remain available.

Core review categories

- Legacy discovery and naming: NetBIOS can reveal host and service information that is no longer needed in DNS-based environments.
- Unnecessary listening services: A service that is listening may become reachable if a firewall rule, VPN path, or network ACL changes later.
- Development and AI test exposure: Local servers launched by coding agents, notebooks, model tools, and demo frameworks are often intended for localhost testing but may bind to all interfaces if misconfigured.
- Business continuity: Hardening should preserve required file sharing, printing, remote support, monitoring, backup, EDR, and administrative access.
- Adjacent protocols: LLMNR and mDNS are not NetBIOS, but they are often reviewed during the same name-resolution hardening effort. Treat them as separate decisions with separate testing.

GreyNOC Research position	Treat NetBIOS and unused ports as cyber hygiene issues. Inventory, test, validate, then enforce through policy or endpoint management.
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04 Before You Make Changes

Preparation prevents outages. Disabling NetBIOS or closing ports can affect name resolution, file shares, printers, remote management, monitoring, and legacy software. Before making changes, collect the current state and create a documented rollback path.

Step	What to do	Why it matters
1	Record current IP settings, adapter names, firewall rules, listening ports, and network profile state.	Creates a rollback point and gives administrators a known-good baseline.
2	Identify required services: file sharing, printing, domain services, RDP, VPN, backup, EDR, monitoring, developer tools, and AI test services.	Prevents accidental outages and separates business-critical ports from unnecessary exposure.
3	Export before-state evidence to a ticket or change record.	Makes the change auditable and easier to troubleshoot later.
4	Test on one device or a small pilot group first.	Confirms impact before broad rollout and limits blast radius.
5	Schedule production changes during a maintenance window.	Reduces business disruption and makes troubleshooting easier.
6	Keep local administrator, console, or out-of-band access available.	Prevents lockout if remote management rules are changed.
7	Document who approved the change and what success looks like.	Improves accountability and makes the release auditable.

Recommended success criteria include: users can sign in, drive mappings still work, printing still works, remote support still works, EDR and backup tools remain healthy, required development workflows still function, and external validation no longer shows ports that were intended to be removed or restricted.

05 Turn Off NetBIOS on a Single Windows Device

Use this method for one workstation or server. You need administrative rights. This is best for single-system testing, troubleshooting, or small environments where settings are not centrally managed.

- Open Control Panel > Network and Internet > Network Connections. You can also press Win + R, type `ncpa.cpl`, and press Enter.
- Right-click the active adapter, then select Properties.
- Select Internet Protocol Version 4 (TCP/IPv4), then select Properties.
- Select Advanced, then open the WINS tab.
- Choose Disable NetBIOS over TCP/IP. Select OK to close each window.
- Restart the device or disable and re-enable the network adapter, then test normal business functions.

Validation tip

After the change, run `nbtstat -n`. If NetBIOS is disabled for that interface, local NetBIOS names should no longer be registered for that interface.

PowerShell inventory commands

```
Get-NetAdapter | Sort-Object Name | Format-Table Name, Status, LinkSpeed
Get-NetIPConfiguration
nbtstat -n
Get-NetIPInterface | Sort-Object InterfaceAlias | Format-Table InterfaceAlias,AddressFamily,Dhcp,ConnectionState
```

Common troubleshooting notes

- If a legacy application stops resolving names, verify whether it was relying on NetBIOS instead of DNS.
- If mapped drives fail, test by fully qualified domain name and IP address to separate name resolution problems from permission or firewall problems.
- If the device is remote, make sure you have a recovery path before changing adapter or firewall settings.
- Document adapter names exactly. Laptops, docking stations, VPN adapters, and virtual switches can create confusing adapter lists.

06 Turn Off NetBIOS Using DHCP in Managed Networks

For DHCP-managed environments, Microsoft documents a DHCP vendor option approach that can disable NetBIOS over TCP/IP for DHCP clients. This is useful when a network team wants consistent behavior across many endpoints without touching each device manually.

- Open the Windows DHCP console on the DHCP server.
- Create or configure the Microsoft vendor class option for NetBIOS behavior.
- Set the option value to disable NetBIOS over TCP/IP for DHCP clients.
- Apply it to the correct scope, renew client leases, and validate on pilot systems.
- Document which scopes have the setting, when it was applied, and who approved the change.

Important

DHCP-based control only affects DHCP clients. Static IP systems may still need manual, script-based, Group Policy, or endpoint-management configuration.

Do not assume one DHCP scope represents the entire environment. Servers, printers, network appliances, lab systems, development VLANs, and isolated networks often have different addressing models. Include those exceptions in the rollout plan so the final state is accurate.

Managed rollout tips

- Start with an OU, device group, or DHCP scope that has low business impact.
- Record DHCP lease renewal timing. Some clients will not reflect the setting until renewal or adapter reset.
- Use a rollback note that names the exact DHCP scope or policy setting to revert.
- For hybrid environments, decide whether the final enforcement method is DHCP, Group Policy, Intune/MDM, endpoint script, or a combination.

07 Find Listening Ports Before Closing Anything

A port is considered exposed when a local service is listening and the firewall or network path allows traffic to reach it. Start with discovery, then decide whether to stop the service, remove an allow rule, add a block rule, or restrict access by source IP.

Discovery should be performed from both the endpoint and another authorized host. Local commands tell you what the machine believes is listening. External scans show what other systems can actually reach. Both views matter.

Windows discovery commands

```
Get-NetTCPConnection -State Listen | Sort-Object LocalPort | Format-Table LocalAddress,LocalPort,OwningProcess
Get-Process -Id <PID>
Get-CimInstance Win32_Service | Where-Object ProcessId -eq <PID> | Select-Object Name,DisplayName,State,StartMode
netstat -ano | findstr LISTENING
```

Firewall review commands

```
Get-NetFirewallProfile | Format-Table Name, Enabled, DefaultInboundAction, DefaultOutboundAction
Get-NetFirewallRule -Enabled True -Direction Inbound | Where-Object Action -eq Allow | Select-Object DisplayName, Profile, Action
Get-NetFirewallRule -Direction Inbound | Get-NetFirewallPortFilter | Where-Object LocalPort -eq <PORT>
```

Finding	Preferred action
Service is not needed	Stop and disable the service, then remove unnecessary allow rules.
Service is needed only by administrators	Restrict by VPN, management subnet, jump box, or source IP.
Service is needed internally but not broadly	Limit scope to required hosts or VLANs.
Service is unknown	Investigate process owner, application dependency, and recent change history before blocking.

08 Close Unused Ports Safely

The best fix is usually to stop or remove the service that is listening. Firewall blocking is helpful, but a disabled service reduces risk more completely because the service is no longer running and cannot be accidentally exposed later by a rule change.

- Stop the service if the application is no longer needed.
- Disable or remove firewall allow rules that no longer have a business need.
- Add a block rule when a service must remain installed but should not be reachable.
- Restrict by source IP for ports that must remain open for administration, monitoring, or approved testing.
- Retest externally from another authorized host or scanner after every change.

Example: block NetBIOS-related inbound traffic on one Windows device

```
New-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS UDP 137" -Direction Inbound -Action Block -Protocol UDP -LocalPort 137
New-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS UDP 138" -Direction Inbound -Action Block -Protocol UDP -LocalPort 138
New-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS TCP 139" -Direction Inbound -Action Block -Protocol TCP -LocalPort 139
```

Example: allow only a management subnet to reach an approved port

```
New-NetFirewallRule -DisplayName "GreyNOC Allow Admin TCP <PORT> from Mgmt" -Direction Inbound -Action Allow -Protocol TCP -LocalPort <PORT>
```

Safer alternative

When a service is needed locally only, reconfigure the application to bind to 127.0.0.1 or localhost instead of 0.0.0.0, then confirm it is not reachable from another host.

09 Ports That Deserve Extra Review

The ports below are commonly reviewed during Windows and small business hardening. The list is not a universal block list. Treat it as a review list and validate each port against business requirements.

Port	Common use	Hardening note
UDP 137	NetBIOS Name Service	Disable NetBIOS if not needed; block inbound where appropriate.
UDP 138	NetBIOS Datagram Service	Usually not needed in modern DNS-based environments.
TCP 139	NetBIOS Session Service	Review with file sharing and legacy application dependencies.
TCP 445	SMB	Do not expose to the internet. Restrict internally to required hosts. Review SMBv1 separately.
TCP 3389	Remote Desktop	Restrict by VPN, source IP, or gateway; avoid broad exposure and require MFA-backed remote access where available.
TCP 5985/5986	WinRM	Restrict to management networks and authorized administrators.
TCP 5900/6080	VNC / web remote desktop	Avoid broad exposure; require VPN, strong authentication, and source restrictions.
UDP 5355	LLMNR	Not NetBIOS, but often reviewed in name-resolution hardening. Test before disabling.
UDP 5353	mDNS	Useful for some discovery workflows. Restrict or segment when not needed.

For small businesses, the highest-risk pattern is broad exposure of management or file-sharing services. If a service must remain available, use layered controls such as VPN, MFA-backed gateway access, source restrictions, endpoint firewall rules, monitoring, and alerting.

10 AI-Assisted Development and Software Test Ports

AI coding agents, test runners, local model tools, notebooks, and prototype frameworks often start temporary web servers or APIs so software can be previewed, tested, or inspected. These services are usually meant for local development. Risk increases when they bind to all interfaces, run without authentication, expose debug features, or remain active after testing is complete.

v1.3 guidance Port 8000 and the related ports in this section should be reviewed on developer workstations, lab servers, build machines, AI test boxes, demo systems, and any host where coding assistants or local model tools run software during testing. Add an expiration or cleanup step for temporary exposure.		
Port	Often associated with	Hardening note
TCP 8000	Python HTTP server, Django/FastAPI/Uvicorn, local API preview, AI bot test harnesses	Common local test port. Bind to localhost unless intentionally shared; block inbound from untrusted networks.
TCP 3000/3001	React, Next.js, Node.js dev servers, Grafana in some environments	Do not leave preview apps broadly reachable. Restrict to localhost, VPN, or approved test subnet.
TCP 5000/5001	Flask, local APIs, development callbacks	Review for unauthenticated debug endpoints and accidental all-interface binding.
TCP 5173	Vite front-end development server	Common with modern web test workflows. Restrict to localhost unless team testing requires access.
TCP 8080/8081	Alternate HTTP, proxies, app servers, test dashboards	Frequently reused. Validate process owner before blocking.
TCP 8888	Jupyter Notebook / JupyterLab	Protect with token/authentication; avoid LAN or internet exposure.
TCP 7860	Gradio demos and AI model user interfaces	Useful for demos, risky if public without authentication. Restrict or tunnel securely.
TCP 8501	Streamlit apps	Often used for quick data/AI apps. Do not expose sensitive data or unauthenticated dashboards.
TCP 11434	Local LLM API services such as Ollama-style endpoints	Treat as local model API exposure. Bind locally and restrict inbound access.
TCP 6333	Vector database APIs such as Qdrant-style services	Protect data stores; do not expose embeddings or collections broadly.
TCP 6006	TensorBoard and training dashboards	Restrict to localhost/VPN; dashboards may reveal model, data, or path details.
TCP 9222	Browser remote debugging protocol	High-risk if reachable. Keep local only and close when testing ends.

11 AI/Test Port Review Workflow

Use this workflow when an AI bot, coding assistant, or automated test tool launches software for preview or validation. The goal is to allow useful testing while preventing accidental exposure.

Step	Action	Expected result
1	List listening ports before starting the bot or test harness.	Baseline shows what was already open.
2	Start the AI-assisted test, dev server, notebook, or local model tool.	New listening ports can be identified.
3	Run local inventory commands and map each port to a process owner.	Each service has a known executable, user, and purpose.
4	Confirm whether the service binds to 127.0.0.1, localhost, a private interface, or 0.0.0.0.	Accidental network exposure is detected.
5	From another authorized host, test whether the service is reachable.	Actual network exposure is confirmed.
6	Stop unused test services and block or restrict anything that must remain running.	The final state matches approved test need.
7	Add cleanup: close terminal sessions, stop containers, remove test firewall rules, and document any exception.	Temporary exposure does not become permanent.

Helpful review commands

```
Get-NetTCPConnection -State Listen | Sort-Object LocalPort | Format-Table LocalAddress,LocalPort,OwningProcess
Get-Process -Id <PID>
netstat -ano | findstr ":8000 :3000 :5000 :5173 :8080 :8888 :7860 :8501 :11434 :6333 :6006 :9222"
```

Decision rule

If a test service is only needed by the person running the test, bind it to localhost. If teammates need access, restrict by VPN, source IP, or test VLAN and add an expiration or cleanup step.

Additional ports to review depending on the stack include TCP 5432 for PostgreSQL, TCP 3306 for MySQL/MariaDB, TCP 27017 for MongoDB, TCP 6379 for Redis, TCP 19530 for Milvus, TCP 9090 for Prometheus, and any custom app port defined by the test harness. These are not AI-only ports, but they often appear in AI application prototypes and should not be exposed casually.

12 Quick Wins, Tips, and Common Mistakes

This section adds practical field notes that help administrators harden safely without turning a simple cleanup into an outage.

Quick wins

Tip	Why it helps
Export before and after states to CSV or TXT.	Creates quick evidence for tickets, customer reports, and rollback.
Name firewall rules with owner, purpose, and review date.	Future administrators can tell why the rule exists.
Prefer RemoteAddress restrictions over broad local-port allows.	A required service can stay available without being reachable by every host.
Block Public profile exposure first for laptops.	Laptops often move between trusted and untrusted networks.
Bind dev services to localhost by default.	Prevents accidental LAN exposure during AI-assisted tests.
Use a short exception window for demos.	Temporary demos are less likely to become forgotten exposure.

Useful copy/paste patterns

```
# Save a quick listening-port baseline
Get-NetTCPConnection -State Listen | Sort-Object LocalPort | Export-Csv .\greynoc-listening-before.csv -NoTypeInformation

# Find process and service from a PID
Get-Process -Id <PID>
Get-CimInstance Win32_Service | Where-Object ProcessId -eq <PID> | Select Name,DisplayName,State,StartMode

# Localhost-only examples
python -m http.server 8000 --bind 127.0.0.1
uvicorn app:app --host 127.0.0.1 --port 8000
jupyter lab --ip=127.0.0.1 --no-browser
```

Common mistakes to avoid

- Treating TCP 445 as NetBIOS. SMB over TCP 445 is related to Windows file sharing review, but it is not NetBIOS over TCP/IP.
- Blocking RDP or WinRM without console access. This can lock out administrators.
- Assuming local LISTENING means externally reachable. Verify from another authorized host.
- Assuming a port number identifies the application. Always map the port to a process owner and service.
- Leaving demo apps, notebooks, tunnels, containers, or browser debug ports running after a test.
- Making a permanent allow rule for a temporary troubleshooting session.

13 Validation, Rollback, and Release Checklist

Validation proves the hardening worked without breaking operations. Rollback proves the team can recover if a dependency was missed. Both should be documented before the change is considered complete.

- Confirm users can still sign in, map drives, print, reach required applications, and connect to management tools.
- Confirm required AI/dev/test workflows still run locally without exposing services broadly.
- From another machine, scan only systems you own or are authorized to test.
- Review Windows Firewall logs, event logs, EDR alerts, and developer tool output after rollout.
- Rollback by re-enabling the affected firewall rule, removing the block rule, or changing the adapter setting back to DHCP/default NetBIOS behavior.
- Record the final state, exceptions, screenshots or command output, and follow-up tasks.

Remove the example NetBIOS block rules

```
Remove-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS UDP 137"  
Remove-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS UDP 138"  
Remove-NetFirewallRule -DisplayName "GreyNOC Block NetBIOS TCP 139"
```

Release item	Complete
Pilot system tested and business functions confirmed	☐
Required services and AI/dev/test exceptions documented	☐
Firewall rules reviewed	☐
RemoteAddress/source restrictions validated for required services	☐
External validation completed from authorized scanner or host	☐
Rollback steps documented and tested where practical	☐
Temporary demo/test exceptions given owner and expiration date	☐
Public-release review completed by document owner	☐

14 Public-Use Disclaimer and Publication Notes

This guide is provided for general cyber hygiene education. Network environments vary. Before applying changes, confirm authorization, test in a pilot group, document rollback steps, and validate that required business services remain available.

GreyNOC is not responsible for outages, misconfigurations, or unauthorized testing resulting from improper use of this guide. Use these recommendations only on systems you own, administer, or are explicitly authorized to assess.

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Distribution	Public after final approval from the approved GreyNOC publication channel.
Release package	Archive source file, final PDF, approval notes, and checksum in the document record.
Release approval	Approval should be recorded in the internal document record before publication. Do not publish from an unapproved personal channel.

Bumped copy note This v1.3 file is a bumped working release created from v1.2. Perform owner review before publishing as the final public release.

Appendix A - References

The following public references support Windows firewall and NetBIOS configuration review. Validate against the current vendor documentation before production deployment.

- Microsoft Learn, "Disable NetBIOS over TCP/IP by using DHCP" - <https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/disable-netbios-tcp-ip-using-dhcp>
- Microsoft Open Specifications, "Vendor-Specific Option Code 0x01 - Microsoft Disable NetBIOS Option" - https://learn.microsoft.com/en-us/openspecs/windows_protocols/ms-dhcpe/ef7676b1-5568-4afc-836a-7eca63a10a3a
- Microsoft Learn, "Windows Firewall rules" - <https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/rules>
- Microsoft Learn, "Configure Firewall Rules with Group Policy" - <https://learn.microsoft.com/en-us/windows/security/operating-system-security/network-security/windows-firewall/configure>
- Microsoft Learn, "New-NetFirewallRule" - <https://learn.microsoft.com/en-us/powershell/module/netsecurity/new-netfirewallrule>
- CISA, "Disable Remote Desktop Protocol (RDP)" - <https://www.cisa.gov/eviction-strategies-tool/info-countermeasures/CM0025>
- NSA / CISA / partners, "Guide to Securing Remote Access Software" - <https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/3417904/nsa-and-co-authors-recommend-best-practices-to-secure-remote-access-software/>

Final note

For public documents, remove draft notes, internal-only references, hidden comments, sensitive diagrams, credentials, live IP addresses, and customer-specific data before release.